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Causal Discoveries for High Dimensional Mixed Data

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Summary: Causal relationships are of crucial importance for biological and medical research. Algorithms have been proposed for causal structure learning among variables of interest. They are able to identify true structures consistently with graphical visualization. While these algorithms have been successfully applied in biological studies where data often follow the same type of distribution, for example, the normal distribution, challenges emerge from epidemiological and clinical studies where data are often mixed with continuous, binary and ordinal variables. We propose to implement a mixed latent Gaussian copula model to estimate the underlying correlation structure via the rank correlations for mixed variables. This correlation structure is then incorporated into a popular causal discovery algorithm, the PC algorithm, to identify causal structures. The proposed algorithm, called the latent-PC algorithm, is able to discover the true causal structure consistently under mild conditions in high dimensional settings. Numerically, the latent-PC algorithm has a much higher true positive rate at the cost of a small increase in the false positive rate, compared with other variants of the PC algorithms. We also apply the latent-PC algorithm in a real data set for hepatocellular carcinoma and provide clinically interpretable causal structures.

Key words: Causal discovery; Directed Acyclic Graph; Mixed data; PC algorithm.

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1. Introduction

It is of crucial importance to understand and identify causal relationships in various scientific disciplines, including biological and medical research. Due to the advancement of biomedical technology, researchers are able to investigate causal relations among gene expression (Spirtes et al., 2000) and signaling pathways (Harris and Drton, 2013). In the medical field, identifying the underlying cause-effect structure allows researchers to further understand the disease mechanism and implement the appropriate intervention and strategy for drug development. For example, understanding of the inflammation process plays an important role in predicting risk of future cardiovascular events (Willerson and Ridker, 2004) and for cardiovascular medicines (Ridker et al., 2017). In immunology, the context of inflammation and its role during physiological immune responses and pathology is constantly evolving (Editorial, 2017). In oncology, where targeted therapies and precision medicines show great success, new biomarkers have been identified using causal models as novel regulators of cell cycle progression and survival in cancer cells (Gendelman et al., 2017). Therefore it is necessary for researches to gain a deep understanding on the underlying causal structures for disease mechanisms.

One of the most popular tools to describe and model causal structures is the directed acyclic graph (DAG). In this graph, nodes represent variables of interest, and edges between nodes depict causal relationships. Moreover, these edges are directed from causes to effects and there is no directed cycle. Estimating the structure of DAGs is in general a difficult task due to the enormous number of possible DAGs. As pointed out by Robinson (1977), the number of possible DAGs is super exponential in the number of nodes.

Most causal structural learning algorithms fall into two categories, score-based methods (Chickering, 2002) where a score criterion is optimized, and constraint-based methods (Spirtes et al., 2000) which implement (conditional) independence tests to infer causal

structures. The past decade has witnessed a number of successful extensions and applications for these two methods. To list a few examples, Kalisch and Bühlmann (2007) proposed estimating DAGs with the PC algorithm for high dimensional Gaussian data. Harris and Drton (2013) relaxed the joint normal assumption in the PC algorithm and proposed the rank-PC algorithm, which also performs well on continuous non-Gaussian data. Ramsey et al. (2017) studied the fast greedy equivalence search algorithm, which greatly improves the computation efficiency of the score based methods. Hybrid methods which combine the strength of score- and constraint-based methods are also widely studied (Tsamardinos et al., 2006; Nandy et al., 2018). For a comprehensive review, see Glymour et al. (2019) and references therein. However, most of these algorithms are designed for one type of data (either continuous or discrete) and are not suitable for mixed data. This severely limits the application of causal DAGs in epidemiological and clinical research where mixed data are often collected. For example, demographic information includes age and gender as continuous and binary variables, respectively. Disease status may be categorized as an ordinal variable with healthy, mild, moderate and severe categories.

Recently the attention in the field is more focused on extending causal discovery algorithms to a mixture of continuous and discrete data. Cui et al. (2016) proposed the Copula-PC algorithm based on a Gaussian copula model. But the algorithm requires Gibbs sampling to estimate the correlation matrix, which is computational expensive and infeasible under high dimensions. Tsagris et al. (2018) implemented the likelihood ratio test to detect the conditional independence under the strong assumption of generalized linear models. However, the validity of the likelihood ratio test is unclear and the algorithm lacks theoretical guarantees to find the true relationships consistently. Andrews et al. (2019) proposed a score-based method for mixed data, but assumed the continuous data to be normally distributed. See Raghu et al. (2018) for a numerical comparison of these algorithms.

While causal discovery algorithms focus on causal structures, another popular method is the graphical model whose goal is to learn the association structure via estimating covariance and precision matrices (Lauritzen, 1996). Due to its mathematical simplicity and wide applicability, Gaussian graphical models have been extensively studied for both estimation (Meinshausen et al., 2006; Yuan and Lin, 2007; Friedman et al., 2008) and statistical inference (Ren et al., 2015; Jankova et al., 2015). We refer interested readers to Drton and Maathuis (2017) for a more comprehensive review. Numerous efforts have been devoted to relaxing the Gaussian assumption (Liu et al., 2009; Xue and Zou, 2012; Li and Solea, 2018). Here we focus on the line of work from Liu et al. (2009), which proposed to model Gaussian and non-Gaussian continuous data by latent variables via the Gaussian copula distribution. Fan et al. (2017) further extended this model to accommodate both binary and continuous data, where binary variables were assumed to be dichotomized from a latent variable following the Gaussian copula distribution. Feng and Ning (2019) showed that this model also works for ordinal data when the latent variable was dichotomized at multiple cut points.

In this paper, we adopt and generalize the latent Gaussian copula model (Liu et al., 2009; Fan et al., 2017; Feng and Ning, 2019) to learn the correlation structure in high dimensional mixed data, which include continuous, binary and ordinal variables. Since correlations directly from mixed data are not suitable, we calculate rank correlations from mixed data and then derive correlations for variables following the latent copula model. We incorporate this correlation structure to the PC algorithm and propose the latent-PC algorithm to identify causal structures. The proposed algorithm includes the PC algorithm (Kalisch and Bühlmann, 2007) and the rank-PC algorithm (Harris and Drton, 2013) as special cases. Under mild conditions, the latent-PC algorithm can correctly identify the true causal structure in high dimensional settings. Our simulation studies show that the latent-PC algorithm shows competitive performance compared to the other variants of PC algorithms.

This paper is organized as follows. We first give a brief introduction of the background of causal structures and the PC algorithm in Section 2.1. Then we introduce the latent copula model and its estimation in Section 2.2 and 2.3, respectively. We conduct extensively simulations in Section 3 and illustrate the advantage of the proposed algorithm using a real data set in Section 4. We conclude our paper with some remarks in Section 5.

2. Methodology

2.1 *Preliminaries and PC Algorithm*

Consider a directed acyclic graph (DAG), denoted as $G = (V, E)$, where the set of nodes V corresponds to a vector of variables $\mathbf{X} = \{X_1, X_2, \ldots, X_d\}^\top \in \mathbb{R}^d$, and the set of directed edges $E \in V \times V$ does not form any directed cycle. A directed edge from X_i to X_j represents the relationship that X_i is a direct cause of X_j for any $1 \leqslant i \neq j \leqslant d$. A *path* in G is a sequence of at least two distinct nodes $X_1, X_2 \ldots, X_k$ connected by edges: $X_i \to X_{i+1}$, $1 \leqslant i \leqslant k-1$. If $X_i \to X_{i+1}$ for all edges, we say that X_k is a *descendant* of X_1. A path between X_i and X_j is *blocked* by a set S if there is a node X_j such that either

(1) $X_i \to X_k \to X_j$ or $X_i \leftarrow X_k \leftarrow X_j$, and $X_k \in S$.

(2) Or $X_i \to X_k \leftarrow X_j$, and neither X_k nor its descendants is in S.

We say two nodes are *d-separated* by S if every path between the two nodes is blocked by S. The probability distribution of $\mathbf{X}$, denoted as P, is *faithful* to the DAG G if

$$X_i \text{ and } X_j \text{ are } d\text{-separated by S} \iff X_i \perp\!\!\!\perp X_j \mid S.$$

The faithfulness property connects the d-separation of X_i and X_j in the DAG (the left-hand side) with the conditional independence between X_i and X_j conditioning on S (the right-hand side). This also serves as the basis to identify causal DAGs from data via independence and conditional independence structures. In the sequel, we assume that the faithfulness property is always satisfied.

One of the most popular algorithms to recover DAGs is the PC algorithm (Spirtes et al., 2000). Specifically, the PC algorithm performs independence and conditional independence tests to infer d-separation relationships among nodes. The algorithm is able to recover the DAG up to its *Markov equivalence class*, which includes DAGs that possess the same d-separation relationships. For example, we list three DAGs in Figure 1 which entail the same conditional independence relationship: $X \perp\!\!\!\perp Y \mid Z$. Thus all three DAGs in Figure 1 are *Markov equivalent*. Technically, we use the *completed partially directed acyclic graphs* (CPDAG) to denote and visualize the equivalence classes of DAGs. In the CPDAG, some edges are directed, while some are undirected to indicate that both directions are possible. We use $C(G)$ to denote the CPDAG of the DAG G. Two graphs G and H are *Markov equivalent* if and only if $C(G) = C(H)$. Thus, the feasible goal of the PC algorithm is to infer $C(G)$.

[Figure 1 about here.]

To perform independence and conditional independence tests in the PC algorithm, Kalisch and Bühlmann (2007) assumed the joint normal distribution and used the Pearson correlation and the partial correlation, respectively. Harris and Drton (2013) proposed the rank-PC algorithm, which further extended to non-Gaussian data via the latent Gaussian copula model (Liu et al., 2009). Both algorithms have been shown to be consistent in identifying the true CPDAG under high dimensional settings. The rank-PC algorithm covers a large class of continuous distributions, but is still unable to handle discrete data or a mixture of continuous and discrete data. Our proposed algorithm, the latent-PC algorithm, can be viewed as an extension of the PC and the rank-PC algorithms because it is directly applicable for continuous, binary and ordinal data, while at the same time achieves the same high dimensional consistency. Next, we describe the mixed latent copula model, which is the backbone of the proposed algorithm.

2.2 *Mixed Latent Copula Model*

In this section, we describe the mixed latent copula model which is built on the Gaussian copula model and its extensions. We first give the formal definition of the Gaussian copula model.

Definition 1 (Continuous Copula): (Liu et al., 2009; Xue and Zou, 2012) A continuous random vector $\mathbf{X} = \{X_1, \ldots, X_d\}^\top \in \mathbb{R}^d$ is said to follow a Gaussian copula model if there exists a set of monotonically increasing transformations $f = \{f_j\}_{j=1}^d$, such that $\{f_1(X_1), \ldots, f_d(X_d)\}^\top \sim N_d(0, \Sigma)$ with the correlation matrix Σ. We denote as $\mathbf{X} \sim GC(\Sigma, f)$.

The Gaussian copula model relaxes the Gaussian assumption in the data, but instead assumes the joint normality of the monotonically transformed variables. Note that marginally one can always transform a random variable to the normal distribution by simply take $f = \Phi^{-1} \circ F$, where F is the distribution function and Φ^{-1} is the quantile function for the standard normal distribution. Thus this is a very mild condition and can be satisfied by a large family of continuous distributions, even if they are skewed, heavy tailed, or multimodal. For simplicity, we exclude the location parameter 0 in $GC(\Sigma, f)$ because it plays little role in characterizing the correlation structure. Next, we describe the definition for the binary copula model.

Definition 2 (Binary Copula): (Fan et al., 2017) A binary random vector $\mathbf{X} \in \mathbb{R}^d$ is said to follow a binary latent Gaussian copula model if there exists a d-dimensional latent vector $\mathbf{Z} \sim GC(\Sigma, f)$ such that

$$X_j = I(Z_j > C_j) \quad \forall j = 1, \ldots, d,$$

where $I(\cdot)$ is the indicator function and C_j, $j = 1, \ldots, d$, are constants. We denote as $\mathbf{X} \sim BGC(\Sigma, f, \mathbf{C})$ with the latent vector $\mathbf{Z} \sim GC(\Sigma, f)$.

In the binary latent Gaussian copula model, we assume that the binary vector is obtained by

dichotomizing a latent continuous vector $\mathbf{Z}$ at unknown constants $\mathbf{C}$. The distribution of $\mathbf{Z}$ is assumed to follow the Gaussian copula and enjoys the flexibility guaranteed by Definition 1.

Definition 3 (Ordinal Copula): (Feng and Ning, 2019) An ordinal random vector $\mathbf{X} \in \mathbb{R}^d$, where $X_j \in \{0, 1, \ldots, N_j\}$ for all $j = 1, \ldots, d$, is said to follow an ordinal latent Gaussian copula model if there exists a d-dimensional vector $\mathbf{Z} \sim GC(\Sigma, f)$ such that

$$X_j = \sum_{k=1}^{N_j} I(Z_j > C_j^k) \quad \forall j = 1, \ldots, d,$$

where $I(\cdot)$ is the indicator function, $C_j^1 < C_j^2 < \cdots < C_j^{N_j}$ are constants, $j = 1, \ldots, d$, and $\mathbf{C} = \{C_1^1, \ldots, C_1^{N_1}, C_2^1, \ldots, C_d^{N_d}\}$. We denote as $\mathbf{X} \sim OGC(\Sigma, f, \mathbf{C})$ with the latent vector $\mathbf{Z} \sim GC(\Sigma, f)$.

The definition for the ordinal copula is in a similar spirit with the binary copula. The main difference is that each element in the latent vector $\mathbf{Z}$ is dichotomized multiple times for the ordinal copula instead of once for the binary copula. Now we are ready to formally define the mixed latent Gaussian copula model for mixed data.

Definition 4 (Mixed Latent Copula): Consider $\mathbf{X} = (\mathbf{X}_1, \mathbf{X}_2, \mathbf{X}_3)$, where $\mathbf{X}_1 \in \mathbb{R}^{d_1}$ is a continuous random vector, $\mathbf{X}_2 \in \mathbb{R}^{d_2}$ is a binary random vector, and $\mathbf{X}_3 \in \mathbb{R}^{d_3}$ is an ordinal random vector, where d_1, d_2 and d_3 are non-negative integers with $d = d_1 + d_2 + d_3 \geqslant 1$. We say that $\mathbf{X}$ follows a mixed latent copula model, denoted as $\mathbf{X} \sim MLGC(\Sigma, f)$, if all of the conditions below are satisfied

- $\mathbf{X}_1 \sim GC(\Sigma_1, f_1)$,
- $\mathbf{X}_2 \sim BGC(\Sigma_2, f_2, \mathbf{C}_2)$ with the latent vector $\mathbf{Z}_2$,
- $\mathbf{X}_3 \sim OGC(\Sigma_3, f_3, \mathbf{C}_3)$ with the latent vector $\mathbf{Z}_3$,
- $(\mathbf{X}_1, \mathbf{Z}_2, \mathbf{Z}_3) \sim GC(\Sigma, f)$.

Under the mixed latent copula model, $\mathbf{X} \sim MLGC(\Sigma, f)$, all randomness in $\mathbf{X}$ comes from

$\mathbf{X}_1$ and the latent random vector $\mathbf{Z}_2$ and $\mathbf{Z}_3$. Thus the most essential cooking ingredient of the PC algorithm, namely the independence and conditional independence structure, resides in the unknown correlation matrix Σ with Pearson correlations. Next, we discuss how to estimate Σ for jointly normal latent variables by rank correlations from mixed data of continuous, binary and ordinal variables.

2.3 *Estimation of the Correlation Matrix*

In this section, we estimate the correlation matrix $\Sigma = (\sigma_{kl})_{k,l=1}^d$ with a unified rank correlation technique for $\mathbf{X} = (\mathbf{X}_1, \mathbf{X}_2, \mathbf{X}_3) \sim MLGC(\Sigma, f)$ as in Definition 4. For two random variables X_k and X_l, $1 \leqslant k \neq l \leqslant d$, their rank correlation, calculated via the Kendall's τ, is $\mathrm{Cor}\{\mathrm{sgn}(X_k - X_k'), \mathrm{sgn}(X_l - X_l')\}$, where (X_k', X_l') is an independent copy of (X_k, X_l) (Kendall, 1948; Li et al., 2021). When observing data $(x_k^1, x_l^1), \ldots, (x_k^n, x_l^n)$, the Kendall's τ rank correlation τ_{kl} can be estimated by the U-statistic

$$\widehat{\tau}_{kl} = \frac{2}{n(n-1)} \sum_{i<j} \mathrm{sgn}\{(x_k^i - x_k^j)(x_l^i - x_l^j)\}. \tag{1}$$

When both X_k and X_l are continuous variables, the rank correlation of X_k and X_l is the same as the rank correlation between $f_k(X_k)$ and $f_l(X_k)$, where $f_k(\cdot)$ and $f_l(\cdot)$ are monotonically increasing functions, and $f_k(X_k)$, $f_l(X_k)$ are jointly normal (Liu et al., 2009). It is shown that simple trigonometric transformations of the rank correlation is an excellent estimator of the corresponding Pearson correlation for normal data. Specifically, Kendall (1948) derived that

$$\tau_{kl} = \arcsin(\sigma_{kl})2/\pi. \tag{2}$$

Given the estimate of $\widehat{\tau}_{kl}$ in (1), we can solve for $\widehat{\sigma}_{kl}$ in a simple closed form.

When X_k is binary and X_l is continuous, Fan et al. (2017) showed that

$$\tau_{kl} = 4\Phi_2(\Delta_k, 0, \sigma_{kl}/\sqrt{2}) - 2\Phi(\Delta_k), \tag{3}$$

where $\Delta_k = f_k(C_k)$, $\Phi(\cdot)$ is the cumulative distribution function of the standard normal

distribution, and $\Phi_2(\cdot,\cdot,t)$ is the cumulative distribution function of the standard bivariate normal distribution with correlation t. Furthermore, when both X_k and X_l are binary,

$$\tau_{kl} = 2\{\Phi_2(\Delta_k, \Delta_l, \sigma_{kl}) - \Phi(\Delta_k)\Phi(\Delta_l)\}. \tag{4}$$

Note that τ_{kl} can be estimated by (1), and Δ_k can be estimated by $\widehat{\Delta}_k = \Phi^{-1}(1-\bar{x_k})$, where $\bar{x_k}$ is the sample mean of X_k. Thus σ_{kl} can be estimated numerically by plugging in $\widehat{\tau}_{kl}$, $\widehat{\Delta}_k$ and $\widehat{\Delta}_l$ into equations (3) and (4) respectively.

When X_k is an ordinal variable with level N_k, Feng and Ning (2019) proposed to consider the binary components $(X_{k\sim 1}, \ldots, X_{k\sim N_k})$, where $X_{k\sim r} = I(X_k \geqslant r)$, $r = 1, \ldots, N_k$. Assume we observe n samples of X_k and its binary components $\{X_{k\sim r}\}_{r=1}^{N_k}$, $i = 1, \ldots, n$. When X_l is continuous, we first estimate the Pearson correlation between $X_{k\sim r}$ and X_l (denoted as $\widehat{\sigma}_{kl}^r$) using (3). Then we can aggregate the N_k correlations together to get the Pearson correlation between X_k and X_l as

$$\widehat{\sigma}_{kl} = \sum_{r=1}^{N_k} w_k^r \widehat{\sigma}_{kl}^r \tag{5}$$

where w_k^r are non-negative weights with the sum equal to 1. Similarly, when X_l is a binary variable, we first estimate the Pearson correlation between $X_{k\sim r}$ and X_l using (4), and then aggregate the N_k correlations together as in (5). When both X_k and X_l are ordinal variables, we first obtain the binary components for both variables, denoted as $(X_{k\sim 1}, \ldots, X_{k\sim N_k})$ and $(X_{l\sim 1}, \ldots, X_{l\sim N_l})$, respectively. Then we can estimate the pairwise Pearson correlation between $X_{k\sim r}$ and $X_{l\sim s}$ using (4) and denote it as $\widehat{\sigma}_{kl}^{rs}$. Finally, we aggregate them to get the Pearson correlation between X_k and X_l as

$$\widehat{\sigma}_{kl} = \sum_{r=1}^{N_k} \sum_{s=1}^{N_l} w_{kl}^{rs} \widehat{\sigma}_{kl}^{rs}, \tag{6}$$

where w_{kl}^{rs} are non-negative weights with the sum equal to 1. One may simply take uniform weights when aggregating the correlations, or use data adaptive ensemble methods as illustrated by Feng and Ning (2019). For simplicity, we use uniform weights in this paper.

Given the estimated correlation matrix Σ derived from the Kendalls' τ rank correlation, we propose the latent–PC algorithm, which identifies the CPDAGs via Σ as the sufficient statistics for independence and conditional independence tests using, for example, the `pcalg` R package (Kalisch et al., 2012).

2.4 *High Dimensional Consistency of the Latent-PC Algorithm*

In this section, we prove the high dimensional consistency of the latent-PC algorithm that it is able identify the true CPDAG consistently. This consistency relies on the accuracy of independence and conditional independence tests, which further depend on the accurate estimation of Pearson correlations and partial correlations via rank correlations.

In the previous section, we were able to estimate Pearson correlations, i.e., elements in $\widehat{\Sigma}$, by equations (2)-(6). For partial correlations, we need to estimate the precision matrix Ω. For example, consider the partial correlation of (X_1, X_2) given X_S, denoted as $\rho_{12|S}$, where $S \subseteq \{3, 4, \ldots, d\}$. Let $\widetilde{\Sigma}$ denote the submatrix of Σ with the $\{1, 2, S\}$-th rows and columns extracted, and let $\widetilde{\Omega}$ be the inverse of $\widetilde{\Sigma}$. It is widely known that

$$\rho_{12|S} = -\frac{\widetilde{\Omega}_{12}}{\sqrt{\widetilde{\Omega}_{11}\widetilde{\Omega}_{22}}}, \tag{7}$$

where $\widetilde{\Omega}_{ij}$ is the (i, j)-th element in $\widetilde{\Omega}$. Then we are able to reject the null hypothesis of conditional independence using the Fisher's z-transform of $\rho_{12|S}$. In the high dimensional setting, the inverse of $\widetilde{\Sigma}$ may not exist. For computational feasibility, we assume that the graph is sparse and its degree, which is the maximum number of edges connected to any node and denoted as $\deg(G)$, is a fixed constant and does not change with the sample size n and the number of nodes d. This sparsity assumption allows us to simply use the inverse of $\widetilde{\Sigma}$ to estimate $\widetilde{\Omega}$. To quantify the estimation error of partial correlations, we need the following lemma, which formally bounds the maximum estimation error of $\widehat{\Sigma}$.

Lemma 1: *For any $k, l, j = 1, \ldots, d$ with $k \neq l$, assume $|\sigma_{kl}| \leqslant 1 - \delta$, and $\Delta_j = f_j(C_j) <$*

M for constants δ and M. For any $\epsilon > 0$, we have

$$P(\|\widehat{\Sigma} - \Sigma\|_\infty > \epsilon) \leqslant Ad^2 \exp(-Bn\epsilon^2)$$

where A, B are constants involving δ and M but not depending on n or d.

The proof of Lemma 1 follows Theorem 3.1 in Fan et al. (2017) and Proposition 1 in Feng and Ning (2019), and is omitted for simplicity. With the accuracy guarantee in Lemma 1, we are able to infer independence and conditional independence structures with precision using mixed latent copula models. Then we can apply the PC algorithm or its variants such as the stable-PC algorithm (Colombo and Maathuis, 2014) to identify the estimated CPDAG, denoted as $\widehat{C}(G)$. The next and final proof is that the estimated CPDAG is consistent with the true CPDAG.

Let $c_1 = \min\{|\sigma_{kl}| : \sigma_{kl} \neq 0, 1 \leqslant k \neq l \leqslant d\}$ and $c_2 = \min\{|\rho_{kl|S}| : \rho_{kl|S} \neq 0, \text{size}(S) \leqslant \deg(G)\}$. In addition, define $\Sigma_{I,I}$ as the submatrix of the correlation matrix Σ with rows and columns extracted for the set $I \subseteq V$ of nodes. Let $\lambda = \min\{\lambda_{\min}(\Sigma_{I,I}) : I \subseteq V, \text{size}(I) \leqslant \deg(G) + 2\}$, where $\lambda_{\min}(\Sigma_{I,I})$ is the smallest eigenvalue of the correlation submatrix $\Sigma_{I,I}$. We obtain the high dimensional consistency of the proposed algorithm in Theorem 1, which proves that the estimated CPDAG of the proposed latent-PC algorithm is equal to the true CPDAG with probability tending to 1. The key idea of the proof is to use Lemma 1 to guarantee that the Pearson correlations and partial correlations are estimated accurately, which lead to high power of the corresponding independence and conditional independence tests. The details of the proof are give in the Appendix.

THEOREM 1: *Let $\mathbf{X}^1, \mathbf{X}^2, \ldots, \mathbf{X}^n$ be i.i.d observations drawn from $\mathbf{X} \sim MLGC(\Sigma, f)$ that is faithful to a DAG G. Let $q = deg(G) + 2$. Then we have*

$$P(\widehat{C}(G) = C(G)) \geqslant 1 - Ad^2 \exp(-Bnc_0^2)$$

where $c_0 = \min\{\frac{c_1}{2}, \frac{\lambda^2 c_2/2}{(2+c_2/2)q+\lambda q c_2/2}\}$.

Theorem 1 shows that the proposed latent-PC algorithm remains consistent when d grows exponentially as n grows linearly. The consistency also relies on the degree of the graph. Overall, the proposed algorithm is suitable for estimating high dimensional sparse causal graphs.

3. Simulations

In this section, we investigate the performance of the proposed algorithm with numerical simulations. Specifically, the DAG with d nodes are generated in two steps. In the first step, we follow Kalisch and Bühlmann (2007) to generate perfectly normally distributed data. The data is then transformed into mixed data in the second step.

Specifically, we denote the nodes as $\mathbf{X} = (X_1, \ldots, X_d)$ and only allow directed edge from X_i to X_j for $i < j$. The existence of an edge follows a Bernoulli distribution, and we set $\text{pr}(A_{ij} = 1) = s$. Here A is the $d \times d$ adjacency matrix, where A_{ij} is equal to 1 if and only if an directed edge appears from X_i to X_j. The parameter s is used to tune the expected number of descendants in the graph, denoted as $E(N)$. A larger s indicates a dense DAG with more edges and a smaller s indicates a sparse DAG. Given a random draw of $A_{ij} = 1$, we replace A_{ij} with an independent realization from the uniform $U(-0.5, 0.5)$ distribution as a quantification of causal effects.

In the first step, the value of X_1 is randomly sampled from $N(0, 1)$ as

$$\epsilon_1 \sim N(0, 1) \text{ and } X_1 = \epsilon_1.$$

For $j = 1, \cdots, d$, we generate the value of the next variable by the structure equation model

$$\epsilon_j \sim N(0, 1), \text{ and } X_j = \sum_{k=1}^{j-1} A_{jk} X_k + \epsilon_j.$$

In the second step, we generate $\mathbf{X}^{new} \sim MLGC(\Sigma, f)$. Without loss of generality, we let d be the multiples of 3, and generate equal proportions of continuous, binary and ordinal variables. Specifically, let $X_k^{new} = f(X_k) = X_k$, $f(z) = z^3$, when $1 \leqslant k \leqslant d/3$ to generate non-

Gaussian data from the Gaussian copula model; let $X_k^{new} = I(X_k > C_k)$ and $C_k \sim U(-1, 1)$ for $d/3 + 1 \leqslant k \leqslant 2d/3$ to generate binary data from the binary latent Gaussian copula model; let $X_k^{new} = I(X_k > C_{k1}) + I(X_k > C_{k2})$, $C_{k1} \sim U(-1, -0.3)$, and $C_{k2} \sim U(0.3, 1)$ for $2d/3 + 1 \leqslant k \leqslant d$ to generate ordinal data. In terms of the complexity of the DAG, we consider two sparsity levels: $E(N) = 3$ and $E(N) = 6$ for a sparse and a dense DAG, respectively. When conducting independence and conditional independence tests, we use two significance levels $\alpha = 0.01, 0.05$, where a more stringent level leads to a higher chance to accept the null hypothesis of independence and conditional independence, and thus a more sparse DAG.

The proposed method is compared with the three algorithms: the vanilla-PC algorithm (Kalisch and Bühlmann, 2007), the rank-PC algorithm (Harris and Drton, 2013), and the oracle-PC algorithm. The last choice, serving as the ideal but impractical case, assumes the normally distributed data in the first step are fully observed. We report the true positive rate (TPR) and the false positive rate (FPR), which are defined as

$$\begin{aligned} \text{true positive rate} &= \frac{\text{true positives}}{\text{true positives} + \text{false negatives}}, \\ \text{false positive rate} &= \frac{\text{false positives}}{\text{false positives} + \text{true negatives}}, \end{aligned}$$

where a positive means an identified edge and a negative means the absence of edges.

The numerical comparison of the four algorithms are summarized in Table 1 and 2, where the sample size n increases linearly while the number of nodes d increases exponentially. As expected, our proposed algorithm can achieve a more accurate estimation on the sparse graph ($E(N) = 3$) compared to the dense graph ($E(N) = 6$). This echos the previous studies in Kalisch and Bühlmann (2007). While the oracle-PC serves as the best case of the performance of PC-algorithms, the proposed latent-PC algorithm performs much better compared to the rank-PC and vanilla-PC in terms of TPR. For example, when the significance level $\alpha = 0.05$, and $(n, d) = (150, 81)$, the TPR of the latent-PC is more than 12% higher than the rank-

PC and the the vanilla-PC. Note that the FPR of the latent-PC is slightly higher with 0.7% higher than the rank-PC and 0.4% higher than the vanilla-PC. In general, the large increase in TPR with a small increase in FPR for the latent-PC indicates that the proposed algorithm is less conservative and is able to discover more edges compared to the rank-PC and the vanilla-PC. We further illustrate the high dimensional consistency of the latent-PC in Figure 2, where we set $E(N) = 6$ and $\alpha = 0.01$. It is clear that the TPR of the proposed algorithm steadily increases as (n, d) increases, and the FPR steadily decreases. This confirms that the proposed latent-PC algorithm is consistent in high dimensional settings.

[Table 1 about here.]

[Table 2 about here.]

[Figure 2 about here.]

4. Real Data Analysis

In this section, we apply the proposed algorithm to the hepatocellular carcinoma (HCC) data set, which is publicly available from the UCI machine learning repository. The data set was investigated by Santos et al. (2015) to predict survival of HCC patients. There are 165 patients diagnosed with HCC. We only consider variables with less than 5% of missing values, and filter out all the subjects with incomplete observations since missing data is not the primary target of this study. After pre-processing, we obtain a data set with $n = 138$ patients and $d = 28$ variables of interest, which consist of 15 continuous variables, 10 binary variables, and 3 ordinal variables. A detailed overview of these variables are presented in Table **??** in the Appendix, which includes variable names, the corresponding abbreviation name in the DAG, and the variable's type, range, mean (for continuous variables) or mode (for binary and ordinal variables).

We perform causal discovery algorithms on the HCC data set with the latent-PC, rank-PC

and vanilla-PC algorithm, whose estimated CPDAGs are illustrated in Figure 3, Figure 4, and Figure 5, respectively. These illustrate the strength of causal discovery algorithms that their graphical visualization is clear and transparent for interpreting and communicating findings. We select the significance level α for the latent-PC using the StARS method (Liu et al., 2010), which is chosen as 0.07. Then we set $\alpha = 0.07$ for the rank-PC and the vanilla-PC to guarantee that all three algorithms are equally conservative. From the three CPDAGs, the latent-PC algorithm is able to discover more meaningful edges. For example, the latent-PC finds 6 variables that are likely to directly influence the survival class (Class): Liver Metastasis, Performance Status, Haemoglobin, Ascites degree, Encefalopathy degree and Alpha-Fetoprotein, all of which have clinical interpretation for HCC patients (Santos et al., 2015). However, the two other PC algorithms can only detect 4 and 5 variables that are likely to directly influence the survival class, respectively. Thus researchers using the latent-PC are better protected against oversimplification, which could happen with causal discoveries (Krieger and Davey Smith, 2016).

[Figure 3 about here.]

[Figure 4 about here.]

[Figure 5 about here.]

5. Conclusions

In this paper, we proposed a modified PC algorithm, the latent-PC, to perform causal discoveries for high dimensional data. The proposed algorithm enjoys high dimensional consistency and can deal with mixed data including binary, ordinal, Gaussian and non-Gaussian variables. Numerical results show that the latent-PC performs much better compared to the vanilla-PC and the rank-PC for high dimensional mixed data.

The idea we adopted is to use rank correlations, which are available for mixed data, to

estimate Pearson correlations among transformed and latent variables, which are normally distributed. Thus we can apply the estimated Pearson correlations as sufficient statistics to the existing PC algorithm to identify CPDAGs. The idea has a broader application to other causal discovery algorithms, for example, the FCI algorithm (Spirtes et al., 2000). This could be readily implemented using the `pcalg` R package.

While the latent-PC algorithm is applicable to a broad mixture of continuous, binary and ordinal variables, a further development could be to extend to time-to-event variables which are also common in clinical trials. This could be an interesting future direction.

References

Andrews, B., Ramsey, J., and Cooper, G. F. (2019). Learning high-dimensional directed acyclic graphs with mixed data-types. *Proceedings of machine learning research* **104,** 4.

Chickering, D. M. (2002). Optimal structure identification with greedy search. *Journal of Machine Learning Research* **3,** 507–554.

Colombo, D. and Maathuis, M. H. (2014). Order-independent constraint-based causal structure learning. *The Journal of Machine Learning Research* **15,** 3741–3782.

Cui, R., Groot, P., and Heskes, T. (2016). Copula PC algorithm for causal discovery from mixed data. In *Joint European Conference on Machine Learning and Knowledge Discovery in Databases*, pages 377–392. Springer.

Drton, M. and Maathuis, M. H. (2017). Structure learning in graphical modeling. *Annual Review of Statistics and Its Application* **4,** 365–393.

Editorial, N. I. (2017). A current view on inflammation. *Nature Immunology* **18,** 825–825.

Fan, J., Liu, H., Ning, Y., and Zou, H. (2017). High dimensional semiparametric latent graphical model for mixed data. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)* **79,** 405–421.

Feng, H. and Ning, Y. (2019). High-dimensional mixed graphical model with ordinal data:

Parameter estimation and statistical inference. In *The 22nd International Conference on Artificial Intelligence and Statistics*, pages 654–663.

Friedman, J., Hastie, T., and Tibshirani, R. (2008). Sparse inverse covariance estimation with the graphical lasso. *Biostatistics* **9,** 432–441.

Gendelman, R., Xing, H., Mirzoeva, O. K., Sarde, P., Curtis, C., Feiler, H. S., McDonagh, P., Gray, J. W., Khalil, I., and Korn, W. M. (2017). Bayesian network inference modeling identifies trib1 as a novel regulator of cell-cycle progression and survival in cancer cells. *Cancer research* **77,** 1575–1585.

Glymour, C., Zhang, K., and Spirtes, P. (2019). Review of causal discovery methods based on graphical models. *Frontiers in Genetics* **10,** 524.

Harris, N. and Drton, M. (2013). Pc algorithm for nonparanormal graphical models. *The Journal of Machine Learning Research* **14,** 3365–3383.

Jankova, J., Van De Geer, S., et al. (2015). Confidence intervals for high-dimensional inverse covariance estimation. *Electronic Journal of Statistics* **9,** 1205–1229.

Kalisch, M. and Bühlmann, P. (2007). Estimating high-dimensional directed acyclic graphs with the pc-algorithm. *Journal of Machine Learning Research* **8,** 613–636.

Kalisch, M., Mächler, M., Colombo, D., Maathuis, M. H., and Bühlmann, P. (2012). Causal inference using graphical models with the R package pcalg. *Journal of Statistical Software* **47,** 1–26.

Kendall, M. G. (1948). Rank correlation methods. *Griffin* .

Krieger, N. and Davey Smith, G. (2016). The tale wagged by the dag: broadening the scope of causal inference and explanation for epidemiology. *International Journal of Epidemiology* **45,** 1787–1808.

Lauritzen, S. L. (1996). *Graphical models*, volume 17. Clarendon Press.

Li, B. and Solea, E. (2018). A nonparametric graphical model for functional data with

application to brain networks based on fmri. *Journal of the American Statistical Association* **113,** 1637–1655.

Li, Z., Wang, Q., and Li, R. (2021). Central limit theorem for linear spectral statistics of large dimensional kendall's rank correlation matrices and its applications. *Annals of statistics* **accepted,**.

Liu, H., Lafferty, J., and Wasserman, L. (2009). The nonparanormal: Semiparametric estimation of high dimensional undirected graphs. *Journal of Machine Learning Research* **10,** 2295–2328.

Liu, H., Roeder, K., and Wasserman, L. (2010). Stability approach to regularization selection (stars) for high dimensional graphical models. In *Advances in neural information processing systems*, pages 1432–1440.

Meinshausen, N., Bühlmann, P., et al. (2006). High-dimensional graphs and variable selection with the lasso. *The Annals of Statistics* **34,** 1436–1462.

Nandy, P., Hauser, A., Maathuis, M. H., et al. (2018). High-dimensional consistency in score-based and hybrid structure learning. *The Annals of Statistics* **46,** 3151–3183.

Raghu, V. K., Poon, A., and Benos, P. V. (2018). Evaluation of causal structure learning methods on mixed data types. *Proceedings of Machine Learning Research* **92,** 48.

Ramsey, J., Glymour, M., Sanchez-Romero, R., and Glymour, C. (2017). A million variables and more: the fast greedy equivalence search algorithm for learning high-dimensional graphical causal models, with an application to functional magnetic resonance images. *International Journal of Data Science and Analytics* **3,** 121–129.

Ren, Z., Sun, T., Zhang, C.-H., Zhou, H. H., et al. (2015). Asymptotic normality and optimalities in estimation of large gaussian graphical models. *The Annals of Statistics* **43,** 991–1026.

Ridker, P. M., Everett, B. M., Thuren, T., MacFadyen, J. G., Chang, W. H., Ballantyne,

C., Fonseca, F., Nicolau, J., Koenig, W., Anker, S. D., et al. (2017). Antiinflammatory therapy with canakinumab for atherosclerotic disease. *New England Journal of Medicine* **377,** 1119–1131.

Robinson, R. W. (1977). Counting unlabeled acyclic digraphs. *Combinatorial mathematics V* pages 28–43.

Santos, M. S., Abreu, P. H., García-Laencina, P. J., Simão, A., and Carvalho, A. (2015). A new cluster-based oversampling method for improving survival prediction of hepatocellular carcinoma patients. *Journal of Biomedical Informatics* **58,** 49–59.

Spirtes, P., Glymour, C. N., and Scheines, R. (2000). *Causation, prediction, and search.* MIT press.

Tsagris, M., Borboudakis, G., Lagani, V., and Tsamardinos, I. (2018). Constraint-based causal discovery with mixed data. *International Journal of Data Science and Analytics* **6,** 19–30.

Tsamardinos, I., Brown, L. E., and Aliferis, C. F. (2006). The max-min hill-climbing bayesian network structure learning algorithm. *Machine Learning* **65,** 31–78.

Willerson, J. T. and Ridker, P. M. (2004). Inflammation as a cardiovascular risk factor. *Circulation* **109,** II–2.

Xue, L. and Zou, H. (2012). Regularized rank-based estimation of high-dimensional nonparanormal graphical models. *The Annals of Statistics* **40,** 2541–2571.

Yuan, M. and Lin, Y. (2007). Model selection and estimation in the gaussian graphical model. *Biometrika* **94,** 19–35.

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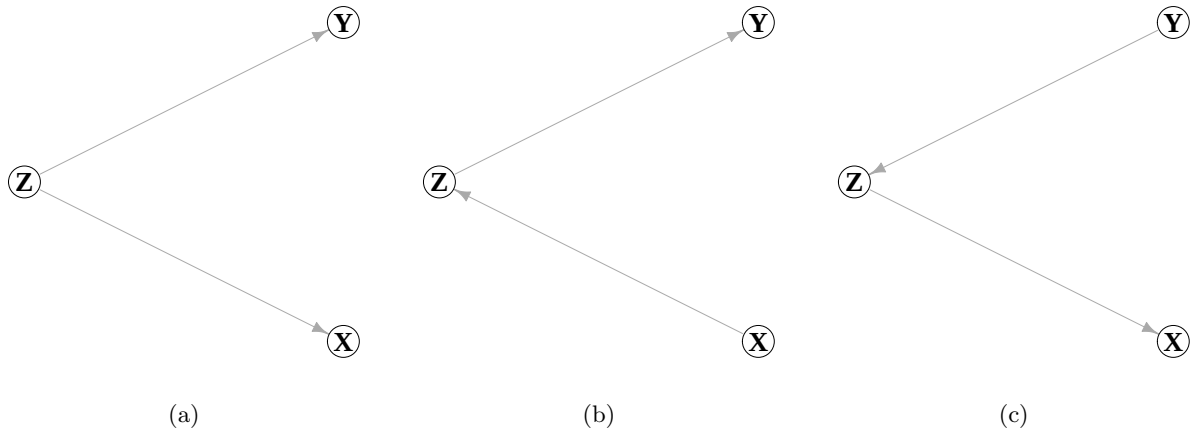
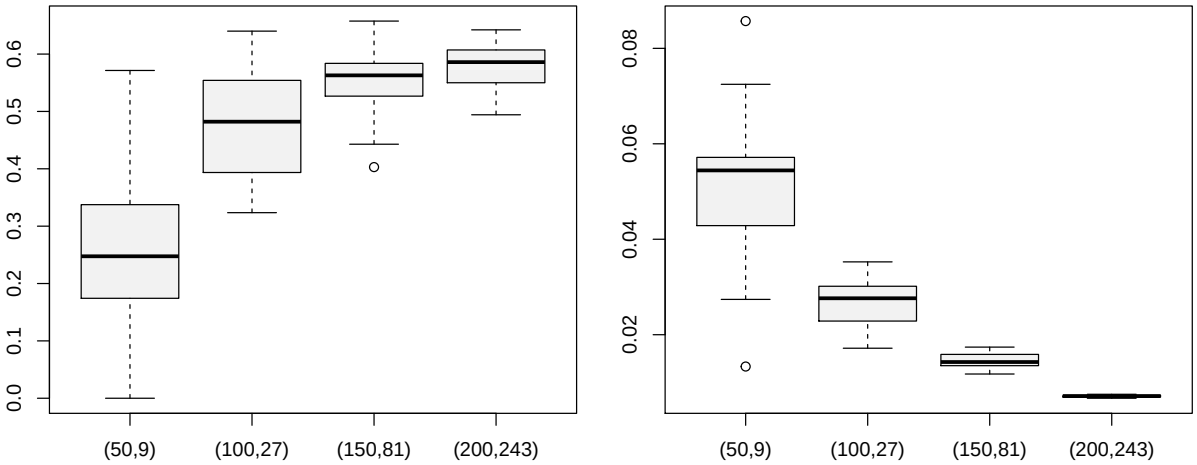


Figure 1: Three different DAGs that entail the same conditional independence relationship $X \perp\!\!\!\perp Y | Z$ and thus they are Markov equivalent.



(a) (b)

Figure 2: For latent-PC: (a) the true positive rate increases while (b) the false positive rate decreases when (n, d) grows linearly and exponentially, respectively.

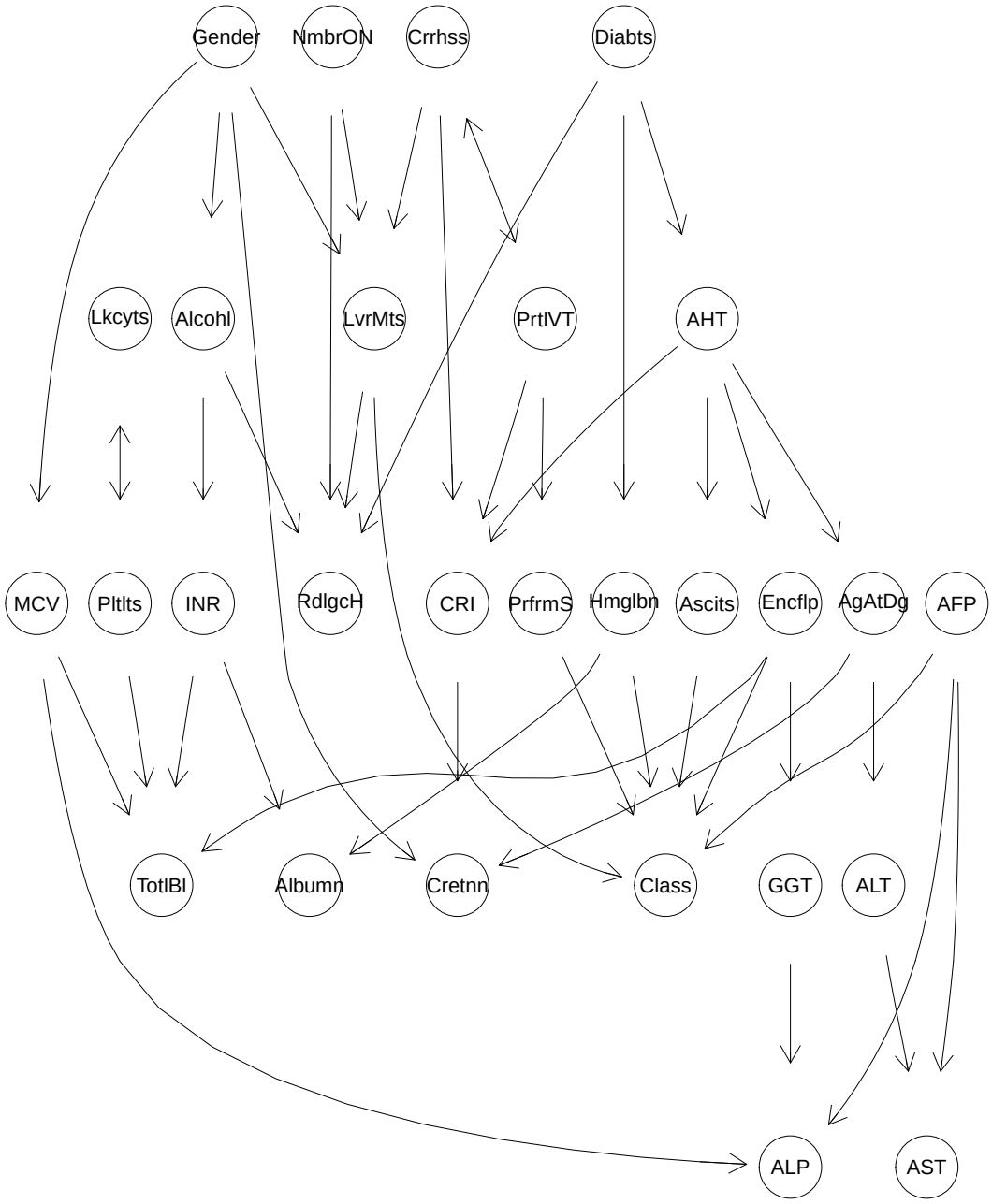


Figure 3: The estimated causal graph from the HCC data set using the latent-PC algorithm.

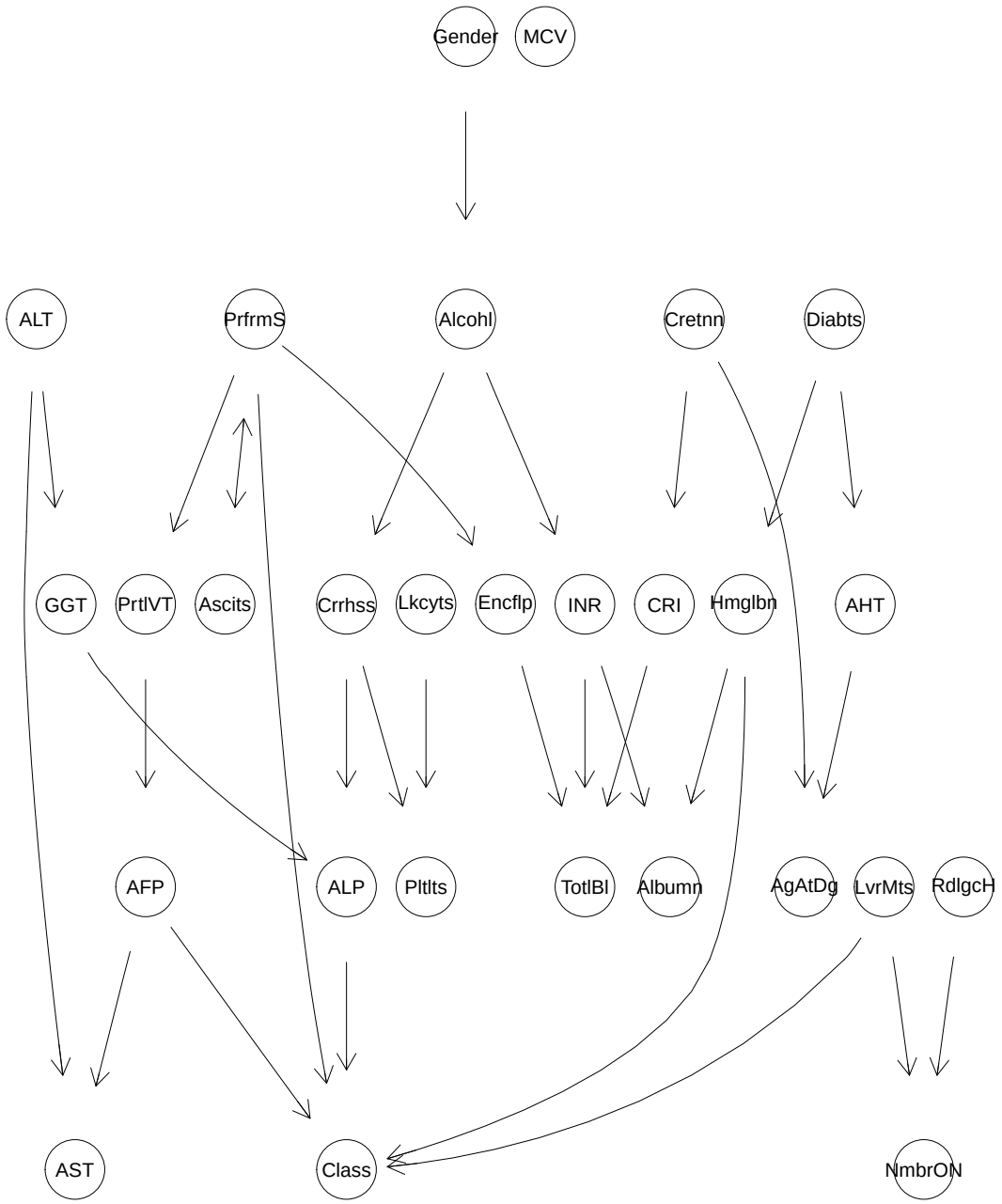


Figure 4: The estimated causal graph from the HCC data set using the rank-PC algorithm.

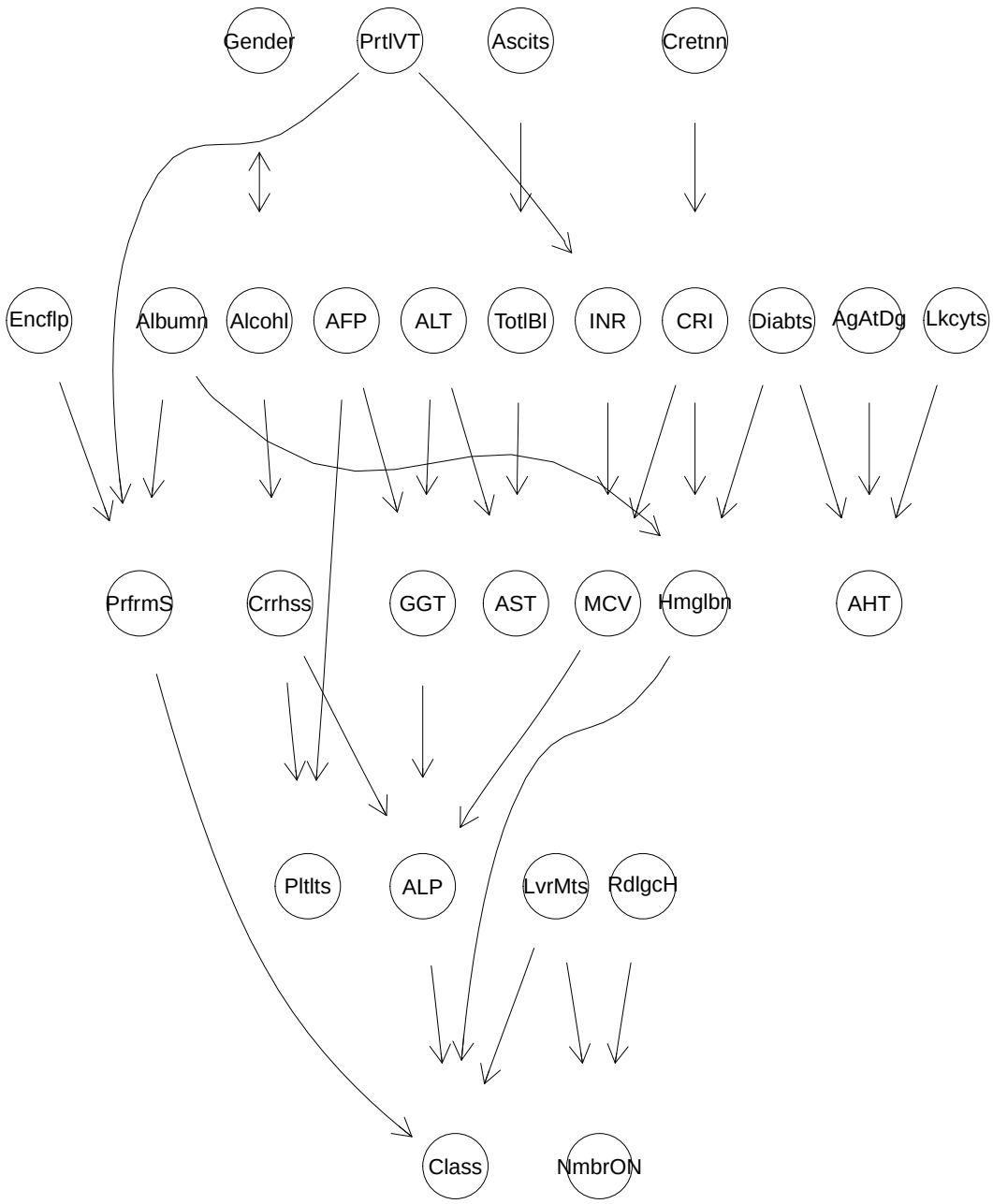


Figure 5: The estimated causal graph from the HCC data set using the vanilla-PC algorithm.

Table 1: The true positive rate and false positive rate of the oracle-PC, latent-PC, rank-PC, and vanilla-PC algorithm when n increases linearly and d increases exponentially. The sparsity level $E(N)$ is set to be 3.

α	(n, d)	Oracle	Latent	Rank	Vanilla	Oracle	Latent	Rank	Vanilla
		True Positive Rate				False Positive Rate			
0.01	$(50, 9)$	0.176	0.182	0.100	0.080	0.034	0.039	0.023	0.026
	$(100, 27)$	0.319	0.310	0.151	0.155	0.010	0.018	0.008	0.010
	$(150, 81)$	0.473	0.417	0.220	0.232	0.003	0.007	0.003	0.004
	$(200, 243)$	0.572	0.470	0.303	0.308	0.001	0.003	0.001	0.002
0.05	$(50, 9)$	0.283	0.322	0.202	0.208	0.041	0.047	0.032	0.034
	$(100, 27)$	0.469	0.404	0.272	0.233	0.013	0.026	0.013	0.017
	$(150, 81)$	0.608	0.528	0.405	0.406	0.006	0.012	0.005	0.008
	$(200, 243)$	0.691	0.557	0.487	0.454	0.003	0.006	0.003	0.005

Table 2: The true positive rate and false positive rate of the Oracle-PC, Latent-PC, Rank-PC, and Vanilla-PC algorithm when n increases linearly and d increases exponentially. The sparsity level $E(N)$ is set to be 6.

α	(n, d)	Oracle	Latent	Rank	Vanilla	Oracle	Latent	Rank	Vanilla
		True Positive Rate				False Positive Rate			
0.01	$(50, 9)$	0.144	0.156	0.059	0.070	0.032	0.037	0.024	0.023
	$(100, 27)$	0.257	0.229	0.127	0.139	0.010	0.014	0.009	0.011
	$(150, 81)$	0.407	0.338	0.193	0.204	0.002	0.005	0.003	0.004
	$(200, 243)$	0.518	0.409	0.275	0.277	0.001	0.002	0.001	0.002
0.05	$(50, 9)$	0.276	0.267	0.158	0.153	0.047	0.046	0.030	0.034
	$(100, 27)$	0.371	0.311	0.222	0.227	0.011	0.021	0.012	0.014
	$(150, 81)$	0.528	0.430	0.335	0.330	0.004	0.009	0.005	0.007
	$(200, 243)$	0.626	0.492	0.428	0.399	0.002	0.005	0.002	0.004